

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.4 Algorithm Execution Observation Table (5 marks)

```
import math
def find_fib(n,c=0):
    c+=1
    if n==0 or n==1:
        return n,c
    l,c=find_fib(n-2,c)
    r,c=find_fib(n-1,c)
    return l+r , c

def generate_execution_observation_table(sizes):
    out=[]
    out.append("Algorithm Execution Observation Table")
    out.append("InputSize RecursiveFactorial IterativeFactorial RecursiveFibonacci
IterativeFibonacci LinearSearch BinarySearch BubbleSort InsertionSort")
    for n in sizes:
        rec_fact=n+1
        iter_fact=n
        fib,rec_fib=find_fib(n)
        iter_fib=n
        lin_search=n
        bin_search=int(math.floor(math.log(n,2))+1)
        bs_sort=int((n*(n-1))/2)
        is_sort=int((n*(n-1))/2)
        out.append(f"{n} {rec_fact} {iter_fact} {rec_fib} {iter_fib} {lin_search} {bin_search}
{bs_sort} {is_sort}")

    return out
```